DSM-Assisted Unsupervised Domain Adaptive Network for Semantic Segmentation of Remote Sensing Imagery

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***Abstract*— The semantic segmentation of high-resolution remote sensing imagery (RSI) is an essential task for many applications. As a promising unsupervised learning method, unsupervised domain adaptation (UDA) methods remarkably contribute to the advancement of high-resolution RSI semantic segmentation. Previous methods focus on reducing the domain shift of orthophotos, suffering from some limitations because the available information in orthophotos is relatively homogeneous. This article proposes a framework to introduce digital surface model (DSM) data for the unsupervised semantic segmentation of RSI. The proposed method combines RSI with DSM through two modules, namely, multipath encoder (MPE) and multitask decoder (MTD), and aligns global data distribution in the source and target domains with a UDA module. A refined postfusion (RPF) module is proposed in the inference phase to exploit the height information fully for refining the segmentation results. Specifically, MPE is designed to utilize RSI and DSM to train the segmentation network jointly, which iteratively fuses RSI and DSM features at multiple levels to enhance their feature representations. MTD is designed to produce fusion prediction maps by filtering interference information of DSM and yielding accurate segmentation masks of DSM and RSI. Experimental results show that the proposed method substantially improves the semantic segmentation performance on high-resolution RSI and outperforms state-of-the-art methods. This article provides a methodological reference for fusing multimodal data in various RSI-based unsupervised tasks.**

***Index Terms*— High-resolution remote sensing imagery (RSI), refined postfusion (RPF), semantic segmentation, unsupervised domain adaptation (UDA).**

NOMENCLATURE

symbol	*meaning*
X_S	RSI set in the source domain.
Y_S	Label set in the source domain.
X_T	RSI set in the target domain.
D_S	DSM set in the source domain.
D_T	DSM set in the target domain.
F_I^n	RSI features from stage n.
F_D^n	DSM features from stage n.
P_I	Segmentation output of RSI.
P_D	Segmentation output of DSM.
P_F	Segmentation output of fusion.

I. INTRODUCTION

THE continuous advances in remote sensing technology and machine equipment are constantly improving the efficiency of obtaining high-resolution remote sensing imagery (RSI). Nowadays, RSI data are becoming increasingly available to the Earth observation community and various applications. In particular, semantic segmentation on RSI is an essential task that converts an RSI into masks to highlight different regions of interest, in which each pixel of the RSI is assigned to a unique category label [1], [2]. The semantic segmentation of RSI is used for applications, including land use and land cover, urban planning, change detection, and urban function extraction [3], [4], [5].

The semantic segmentation methods of RSI are boosted by flourishing supervised deep learning technologies [6], [7]. For example, numerous works, represented by the deep convolutional neural network and its variants, are developed for the semantic segmentation of RSI, achieving considerable performance improvements [8], [9]. Those works mainly follow supervised deep learning approaches to train models, which are highly dependent on massive quantities of manually labeled images. In practice, manually labeling images is labor intensive and time consuming.

RSI semantic segmentation methods based on weakly supervised and unsupervised methods are emerging to reduce the

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substantial cost of manual labeling. As a promising unsupervised learning method, the unsupervised domain adaptation (UDA) strategy has been studied and achieved great performance improvement [10]. UDA-based methods try to transfer the segmentation knowledge of RSIs from the densely well-labeled source domain to the unlabeled target domain. For example, the multiple weakly supervised constraint model was proposed to learn deep semantic segmentation networks for cross-domain RSI semantic segmentation [11]. To narrow the intraclass and interclass differences, a category-aware generative adversarial network was developed for the UDA semantic segmentation of RSI [12]. Considering the inconsistency between different categories and regions [13], a course-style local-to-global UDA segmentation framework was proposed. These works mainly employ single-modal data, orthophotos, to train the model. They suffer from some limitations because the available information in orthophotos is relatively homogeneous. For example, the segmentation models are not very discriminative for some ground objects with similar spectral features or in shaded regions, resulting in misclassification often occurring between these objects, such as buildings and roads, low vegetation, and trees.

To address the issue, this study introduces digital surface model (DSM) data to improve the UDA semantic segmentation of RSIs. DSM data provides the height information of ground objects, which characterizes domain-invariant features for discriminating objects in imageries [14]. Specifically, the proposed method first jointly performs multipath encoding and multitask decoding of DSM and RSI features, then combines the UDA strategy to train the model, and finally refines fusion features to predict ground objects. The original contributions of this article are as follows.

1) This study highlights the role of DSM data in UDA semantic segmentation tasks of RSI, and proposes to improve the segmentation performance by introducing DSM.
2) A domain adaptive segmentation framework that fuses DSM and high-resolution RSI is proposed. The framework designs a multipath encoder (MPE) module that uses dual attention to optimize the feature representation of DSM from low levels to high levels, and a multitask decoder (MTD) module that introduces a filter for focusing local regions of DSM and a gated structure for fusing data from different modalities. The framework provides methodological references for fusing multimodal data to improve the performance of RSI-based tasks.
3) Extensive experiments on the ISPRS 2D Semantic Labeling Contest dataset show that the proposed method is superior to state-of-the-art UDA methods.

The rest of this article is organized as follows. The related works are briefly reviewed in Section II. The proposed method is described in detail in Section III. The experimental results are reported and analyzed in Section IV. Ablation studies as well as some factors that may affect model performance are discussed in Section V. Finally, the conclusions are presented in Section VI.

II. Related Work

A. Supervised Semantic Segmentation for RSI

Benefiting from continuous development in deep learning that can learn model parameters from labeled samples, the supervised approach has been greatly advanced to perform semantic segmentation tasks on RSI. Especially, the first end-to-end full convolutional network [1] is increasingly used in studies that employ convolutional networks to capture the semantic features of RSI and predict semantic segmentation labels [15], [16]. Moreover, approaches have been derived to address the limitations of convolutional neural networks [9], [17], [18]. These works are devoted to enhancing the relation of contexts or improving edge prediction to facilitate the accurate identification of ground objects. For example, ABCNet designed bilateral structures to capture spatial details and global contextual information for improving segmentation accuracy [9]. A multiscale context aggregation network was designed to reduce the loss of spatial information, in which a spatial inference module was used to learn spatial correlations, and an adaptive spatial pool was used to aggregate contexts [19]. E-D-Net designed two subnetworks to capture the edge information and detail information of images, which considerably improved the problem of poor edge prediction in building extraction by integrating these two types of information [20]. Although those supervised methods have gained great progress in the ability of RS semantic segmentation, they highly depend on massive quantities of high-quality labeled images and suffer from the limitations of being labor intensive and time consuming.

B. Unsupervised Semantic Segmentation for RSI

To minimize the dependence on high-quality labels, UDA-based semantic segmentation methods have been extensively studied. Those methods can be classified into four categories, generation-based methods [21], [22], [23], adversarial-based methods [24], [25], entropy-based methods [26], and self-training methods [27], [28], [29]. In particular, the adversarial-based approach is very promising because it does not require explicit modeling of potential objects and can align different domains in pixel-level, feature-level, and output-level spaces. It plays a large part in UDA semantic segmentation and has evolved from global adversarial [25] to category-level adversarial and semantic-level adversarial in a continuous improvement [30], [31]. To improve adversarial strategies, it has been combined with different UDA approaches, such as using translation models to reduce the interdomain gap [32], or adding course learning to minimize the intradomain gap [24].

Inspired by the success of UDA-based semantic segmentation methods on computer vision, studies have started to derive those methods for segmenting RSIs. Remote sensing scholars have proposed various UDA strategies to narrow the gap between domains, between classes, and within classes. These methods have tried to reduce misclassification and confusion between ground objects. For example, to facilitate interdomain and intradomain alignment, Cai et al. [33] proposed an iterative

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intradomain adaptive method to reduce the intradomain gap. A weakly supervised domain adaptive framework was established in combination with adversarial entropy to improve the prediction accuracy of the target [34]. Some studies have further explored the feasibility of aligning data distributions at the category level. Chen et al. [35] developed region and category adaptive domain discriminators to reduce negative migration. Wu et al. [36] explicitly align category features using depth covariance and proposed a covariance regularization method to improve robustness. Xu et al. [12] proposed class-aware generative adversarial networks to reduce intraclass and interclass differences in the source and target domains. Local feature alignment between regions was achieved by learning adaptive weights for the target prediction probability map to measure the differences between regions [37]. A course-based local-to-global cross-domain adaptation framework was developed for aligning features from local semantics to global structure [13]. Generation-based methods, such as translating images and adding multiple weakly supervised constraints to the segmentation network, were used for improving RSI cross-domain segmentation accuracy [11]. Moreover, BiFDANet defined bidirectional semantic consistency based on the idea of bidirectional learning and linearly combined the outputs of the two domains to obtain more reliable prediction results [38]. In addition, several methods are available to solve derived problems. For instance, to reduce the effect of resolution inconsistencies, ResiDualGAN introduced the DualGAN-based residual structure and resizing module [39]. Yan et al. [40] focused on pixels that were not assigned pseudolabels and developed cross-mean-teacher networks to utilize all pixels efficiently. These approaches have facilitated the progress of UDA semantic segmentation tasks for RSI. However, ground objects have similar spectral characteristics but different classes in orthophoto images, in addition to shadow areas caused by illumination. Identifying these areas by relying only on orthophoto images is very prone to errors.

C. Semantic Segmentation With Multimodal Data

Existing work has explored numerous approaches to fuse multimodal data because various remote sensing data may contribute to RSI segmentation. These methods combine various remote sensing data to improve segmentation accuracy [41], [42], such as DSM and normalized difference vegetation index/visible-band difference vegetation index.

Benefiting from DSM can provide the height information to assist in separating ground objects, and a number of studies have deeply explored the feasibility of using DSM to improve the semantic segmentation task of RSI. Zhang et al. [43] proposed an end-to-end model to fuse DSM for improving prediction accuracy. Du et al. [44] employed Dempster–Shafer theory to combine the prediction results of DSM and RSI and made performance improvements by combining various uncertain indicators in classification. DSMFNet investigated the effectiveness of four fusion strategies and introduced DSM for improving model performance in hard-to-classify areas [45]. These methods purposely address the limitations of unimodal data and verify the validity of DSM inputs. Many researchers have studied the fusion method and fusion moment in-depth, which strongly address the heterogeneity of different modality data. For example, GRRNet developed a gated residual refinement network, which introduces a gating structure to reduce unnecessary feature transmission and refine the classification results [46]. DSPCANet designed location and channel attention fusion branches for the DSM branch to learn geographic feature representations and modeled the fusion features efficiently to recalibrate the category information [47]. An end-to-end cross-modal gated fusion network was employed for adaptive feature learning, which uses two independent encoders to extract multilevel features from RGB and DSM data, and weighs each modality to remove irrelevant parts [48]. A hybrid attention-aware fusion network was proposed to learn and utilize individual modal and cross-modal features fully to improve the ability of building extraction [49].

The above-mentioned studies have shown that complementary multimodal data can provide a more accurate basis for segmentation compared with unimodal data. However, a few works have exploited domain-invariant characteristics of DSM to improve UDA-based semantic segmentation on RSI. This article investigates the methods to fuse DSM and RSI fully for UDA semantic segmentation.

III. Methodology

A. Framework

In this article, $S = \{X_S, D_S, Y_S\}$ denotes the labeled source domain, and $T = \{X_T, D_T\}$ denotes the unlabeled target domain. The key notations used in this section are shown in the Nomenclature.

The proposed framework, named MDDANet, which combines DSM and RSI with UDA to improve the accurate recognition capability of multiple categories of ground objects, is illustrated in Fig. 1. The framework consists of four key modules, namely, MPE, MTD, domain adaptation module, and refined postfusion (RPF). MPE encodes and fuses RSI features and DSM features by connecting two independent encoders. The feature representations learned in RSI are adaptively fused into DSM during the extraction of low-level features to high-level features. MTD decodes the feature maps from MPE to acquire purer DSM and finer RSI segmentation outputs and obtains the final segmentation results by adaptive fusion. Domain adaptation facilitates the data distribution of the target domain to be close to the source domain. RPF is designed to refine the fusion segmentation results further by using DSM prediction maps in the inference phase. These four modules are detailed described in Sections III-B–III-E, respectively.

B. Multipath Encoder Module

The MPE module includes two different encoders, namely, ResNet34 and Deeplabv2 with ResNet101 as the backbone, and the encoders are divided into five stages. The module progressively encodes DSM and RSI to fuse the features of both modalities for achieving enhanced feature representation. Inspired by [50], for each stage, an attention-aware fusion (AAFU) block is employed to combine the feature representations learned from the texture and color of RSI with the feature representations learned from the height of

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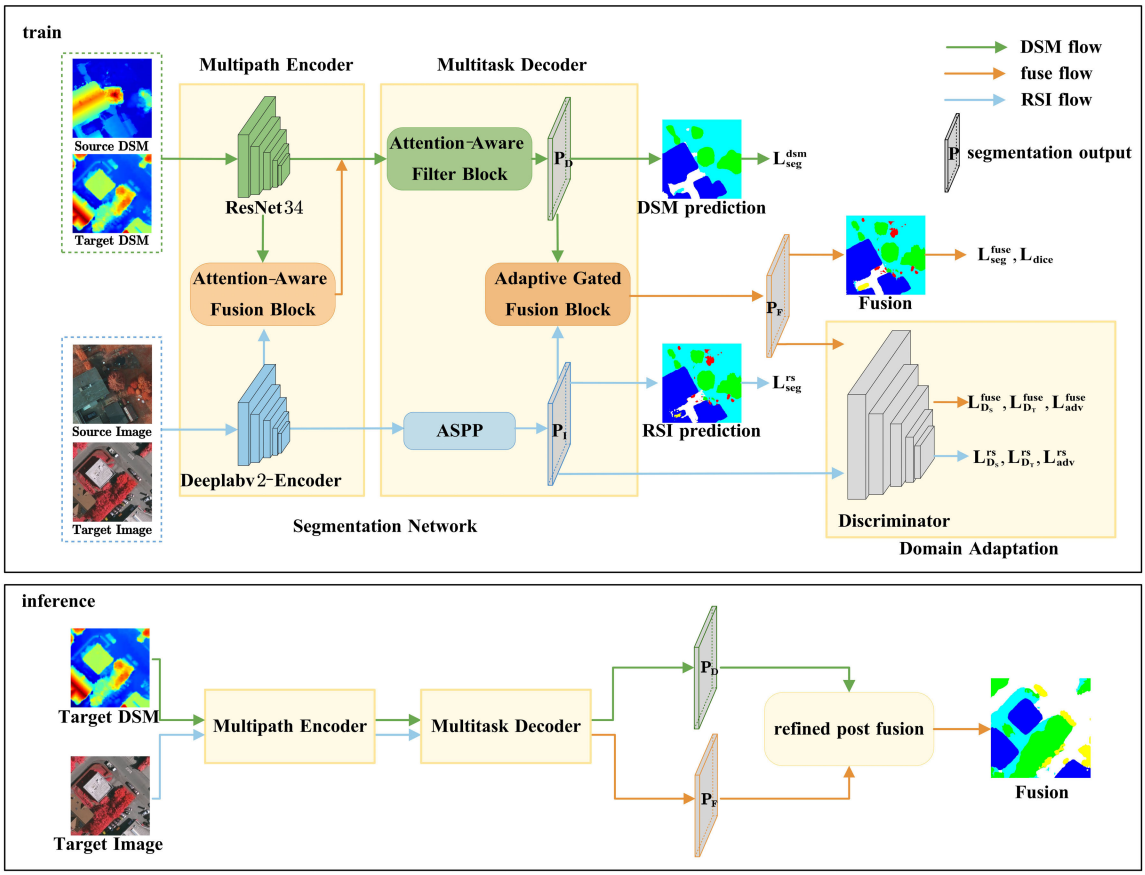


Fig. 1. Overall architecture of multimodal domain adaptive framework (MMDANet).

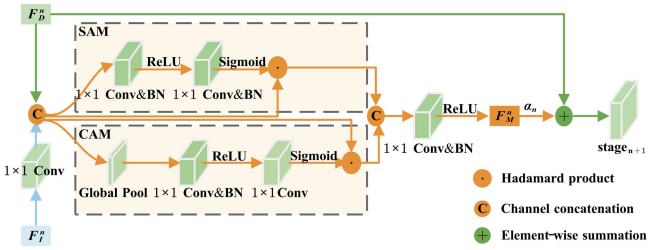


Fig. 2. Specific structure of AAFU block.

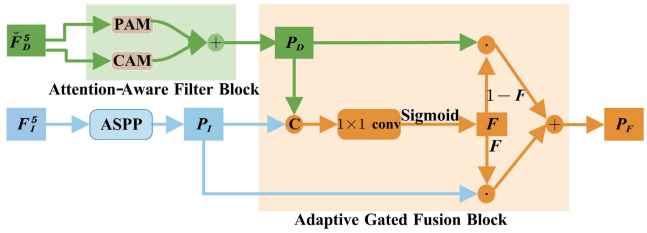


Fig. 3. Structure of MTD module.

DSM. Then, the fusion features are fed into the next stage via the weighted summation of the original DSM features. The weighted summation is used to weaken the fusion features further, which helps reduce the interference information from RSI.

AAFU consists of a channel attention module (CAM) and a spatial attention module (SAM), and its structure is illustrated in Fig. 2. For RSI features F_I^n and DSM features F_D^n obtained in the nth stage, the channel number of F_I^n is four times larger than F_D^n when n is from 2 to 5, and they have the same number of channels when the value of n is 1. First, F_I^n inputs 1×1 convolution to obtain $\widehat{F_I^n}$ for Stages 2–5. Then, F_D^n and $\widehat{F_I^n}$ are concatenated in the channel dimension and fed into the CAM and SAM. The final obtained features F_M^n are multiplied with the adaptive weight α_n and next added to F_D^n again to obtain the new DSM features F_D^n, which are input in the next DSM stage.

C. Multitask Decoder Module

The MTD module learns model parameters through two tasks, RSI prediction and DSM prediction. Fig. 3 shows the module includes an atrous spatial pyramid pooling (ASPP) for decoding RSI features, an attention-aware filter block (AAFI) for focusing on local information of DSM features, and an adaptive gated fusion block for fusing DSM and RSI features.

In the proposed framework, DSM aims to help classify four categories of ground objects that vary in height but are easily confused for classification, including "buildings," "impervious surfaces," "trees," and "low vegetation." Nevertheless, some noise may still be involved in the encoded features by MPE. To focus more on effective feature information, a dual-attention mechanism [51] is employed in AAFI, which includes a position attention module (PAM) and a CAM. In addition, a gated structure [48] is employed in the adaptive gated fusion block that adaptively assigns weights to the

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segmentation outputs of DSM and RSI for obtaining a fused segmentation output.

First, the output features of the encoder of DSM in the last stage are input into AAFI to obtain the segmentation output P_D, and F_I^5 is fed to ASPP to obtain P_I. After up sampling to 512×512, P_D and P_I are connected in the channel dimension and sent into a 1×1 convolution layer to obtain the adaptive weight F, in which F determines the proportion of P_D and P_I retained in the fusion segmentation outputs. P_D and P_I are multiplied by F and $(1 - F)$, respectively, and the obtained results are added to generate the final fusion segmentation outputs P_F.

To train the segmentation network, segmentation loss is used to supervise the learning of the network on the source domain. The standard cross-entropy loss is used to calculate the segmentation loss $L_{\text{seg}}^{\text{rs}}$, $L_{\text{seg}}^{\text{fuse}}$, and $L_{\text{seg}}^{\text{dsm}}$, as expressed in (1)–(3), respectively. The loss requires the input of segmentation outputs and the corresponding true label maps. P_I^S, P_F^S, and P_D^S refer to segmentation outputs of the RSI features, fusion features, and DSM features in the source domain, respectively. To obtain $L_{\text{seg}}^{\text{dsm}}$, the label data adopt a simple filtering mechanism. During the training, C^* contains only four types of labels, that is, the pixels labeled "clutter" and "car" are ignored

$$L_{\text{seg}}^{\text{rs}} = -\sum_{h\in H}\sum_{w\in W}\sum_{c\in C} Y_S \log\left(P_I^S\right) \tag{1}$$

$$L_{\text{seg}}^{\text{fuse}} = -\sum_{h\in H}\sum_{w\in W}\sum_{c\in C} Y_S \log\left(P_F^S\right) \tag{2}$$

$$L_{\text{seg}}^{\text{dsm}} = -\sum_{h\in H}\sum_{w\in W}\sum_{c\in C^*} Y_S \log\left(P_D^S\right) \tag{3}$$

where $L_{\text{seg}}^{\text{rs}}$, $L_{\text{seg}}^{\text{fuse}}$, and $L_{\text{seg}}^{\text{dsm}}$ present the segmentation losses of the RSI features, fusion features, and DSM features, respectively. H, W, and C denote the height, width, and category number of the segmentation output, respectively. Y_S presents the true label.

In addition, dice loss is adopted to constrain the learning of the fusion prediction results further and reduce the effect of class imbalance, as defined in the following:

$$L_{\text{dice}} = 1 - \left(2\left|Y_S \cap P_F^S\right| + \varepsilon\right) / \left(|Y_S| + \left|P_F^S\right| + \varepsilon\right) \tag{4}$$

where ε is set to 1 to ensure that L_{dice} is not an undefined value in the any case. "$\cap$" means intersection.

D. Domain Adaptation Module

The above-mentioned Sections III-B and III-C describe the training of DSM and RSI in the source domain. The proposed framework employs a UDA module that transfers the segmentation network learned under the supervision of the source domain to the target domain. The module aligns the data distributions of the target and source domains to each other via a classical output-level adversarial method. The DSM encoder focuses on learning features of ground objects with height differences and is not engaged in domain adaptation. The domain gap of DSM features can be reduced indirectly by gradient back-propagation.

Under the constraints of adversarial loss and discriminator loss, the segmentation network and discriminator are alternately optimized. The adversarial loss is utilized to judge the gap between the target and source domains, thus guiding the optimization direction of the segmentation network. The adversarial loss function of RSI and fusion features is given in the following:

$$L_{\text{adv}}^{\text{rs}} = -\sum_{(H',W')} \log\left(D\left(\text{softmax}\left(P_I^T\right)\right)\right) \tag{5}$$

$$L_{\text{adv}}^{\text{fuse}} = -\sum_{(H',W')} \log\left(D\left(\text{softmax}\left(P_F^T\right)\right)\right) \tag{6}$$

where $H' = H/32$ and $W' = W/32$. P_I^T and P_F^T refer to the segmentation outputs of RSI features and fusion features in the target domain, respectively. $L_{\text{adv}}^{\text{rs}}$ and $L_{\text{adv}}^{\text{fuse}}$ represent the adversarial losses of P_I^T and P_F^T. The purpose of the discriminator is to identify the origin of the input sample accurately. The discriminator loss is used to maximize the feature difference between the source and target domains, and the discriminator losses are given in (7) and (8). Z is a domain indicator flag. If a sample is from the source domain, Z is set to 1, and 0 otherwise

$$\begin{aligned} L_D^S = &-\sum_{(H',W')} Z \log\left(D\left(\text{softmax}\left(P_I^S\right)\right)\right) \\ &+ Z \log\left(D\left(\text{softmax}\left(P_F^S\right)\right)\right) \end{aligned} \tag{7}$$

$$\begin{aligned} L_D^T = &-\sum_{(H',W')} (1-Z) \log\left(1 - D\left(\text{softmax}\left(P_I^T\right)\right)\right) \\ &+ (1-Z) \log\left(1 - D\left(\text{softmax}\left(P_F^T\right)\right)\right) \end{aligned} \tag{8}$$

where $H' = H/32$, $W' = W/32$. P_I^S and P_F^S refer to RSI and fusion segmentation outputs in the source domain, respectively. L_D^S and L_D^T denote the discriminator loss in the source and target domains, respectively.

E. Refined Postfusion Module

In the inference period, the MTD modules are employed to generate the DSM and fusion segmentation outputs. These outputs are then fed into the SoftMax layer to obtain the probability map. Next, DSM probability maps are refined by a filtering operation to extract information further, which is beneficial for separating ground objects. Specifically, the DSM probability map contains the probability of each pixel being classified into each category. The module selects the pixels where the probability of four categories is greater than the threshold μ to generate the mask W_a. Similarly, the mask W_b is generated by selecting the pixels predicted as not in the "car" category from fusion probability maps. The selection of a category is discussed in Section V-C. The module performs a "logic AND" operation on W_a and W_b to obtain W_c that meets the two conditions.

The W_c and DSM probability maps are multiplied to generate the filtered results. The final result is obtained by directly adding the filtered results and the fusion probability map, as illustrated in Fig. 4.

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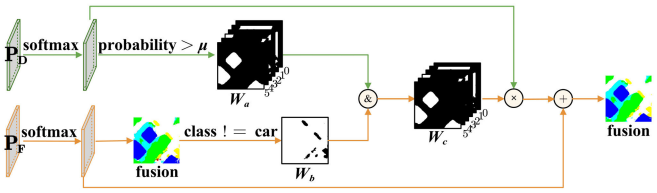


Fig. 4. Flow of RPF module.

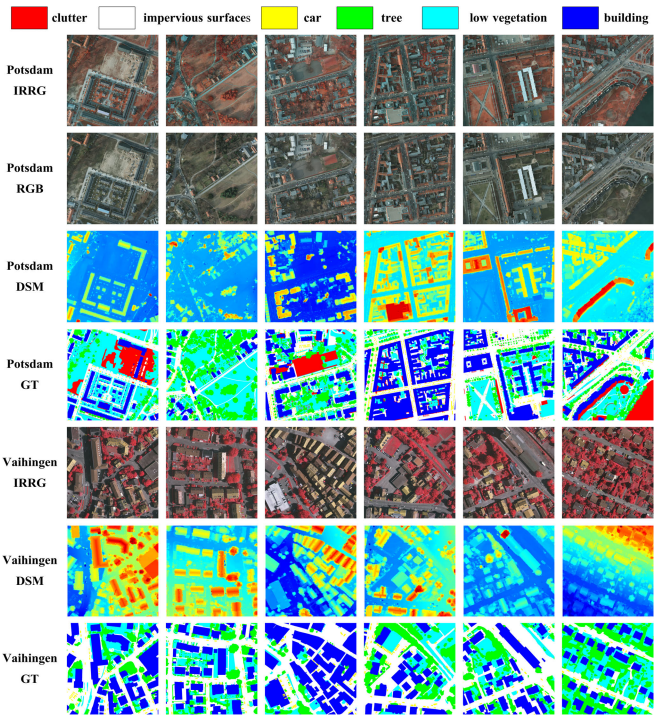


Fig. 5. Some examples of the Potsdam IRRG dataset, Potsdam RGB dataset, and Vaihingen IRRG dataset.

F. Model Optimization

For the training model, the loss function of total semantic segmentation networks is shown in the following:

$$L_G = L_{\text{seg}}^{\text{rs}} + L_{\text{seg}}^{\text{fuse}} + L_{\text{seg}}^{\text{dsm}} + L_{\text{dice}} + L_{\text{adv}}^{\text{rs}} + L_{\text{adv}}^{\text{fuse}}. \quad (9)$$

The discriminator distinguishes whether the image belongs to the source or target domain. Its loss is calculated by the following:

$$L_D = L_D^S + L_D^T. \quad (10)$$

IV. Experiments

A. Datasets

The proposed method is evaluated on the Potsdam and Vaihingen datasets. The two datasets belong to ISPRS 2D Semantic Labeling Contest dataset [57] and contain high-quality labeled RSIs and corresponding DSM data. Several examples are shown in Fig. 5.

The Potsdam dataset consists of 38 high-resolution true orthophotos (TOP) and their DSM. The TOPs of the dataset are available in three channel compositions, namely, red-green-blue (RGB), infrared red-green (IRRG) and red-green-blue infrared (RGB-IR). The IRRG and RGB images are used in this article. The size of all images is fixed at 6000 × 6000. The ground sampling distances of RSI and DSM are 5 cm. The pixels of those images are labeled into one of six categories, namely, "clutter/background," "impervious surfaces," "car," "tree," "low vegetation," and "building." In this experiment, RSI, DSM, and label images are cropped at a fixed step of 512 pixels, obtaining 4598 images with a pixel size of 512 × 512.

The Vaihingen dataset consists of 33 high-resolution RSI and DSM with a size of approximately 2000 × 2000, where the TOPs are available in IRRG mode. The ground sampling distances of RSI and DSM are 9 cm. This dataset has the same label categories as the Potsdam dataset. Similarly, RSI, DSM, and label images in the Vaihingen dataset are cropped with a fixed step size of 256 pixels, obtaining 1698 images of 512 × 512 pixel size.

B. Implementation Details

For the segmentation network, the ResNet34 [52] network is employed to encode the DSM features, the Deeplabv2 [53] framework with ResNet-101 [52] model pretrained on ImageNet [54] is used to encode RSI features, and ASPP [53] is used as a classifier with sampling rate at 6,12,18,24. The stride of the last two convolutional layers of ResNet34 is modified from 2 to 1 to make the output size of both encoders the same.

The discriminator is set up in the same way with AdaptSegNet [25]. Specifically, the network consists of five convolutional layers with 4 × 4 convolutional kernels and a stride of 2, where the number of channels is 64, 128, 256, 512, 1. The output of each convolutional layer is passed through a leaky ReLU layer with a parameter of 0.2.

Our framework is implemented with the PyTorch framework [55], and all experiments are conducted on a workstation with an NVIDIA Tesla T4 GPU and 16 GB of memory. In the training, a stochastic gradient descent optimizer with Nesterov acceleration is used [56], where the momentum is set to 0.9. The weight decay is set to 5×10^{-4}, and the initial learning rate is set to 2.5×10^{-4}. The batch size and epoch are set to 2 and 10, respectively. The training settings of the discriminator are the same as those of AdaptSegNet.

C. Evaluation Metrics

In the experiment, F1_Score and IoU are employed to evaluate the performance of the proposed method. F1_Score is defined as follows:

$$\text{precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (11)$$

$$\text{recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (12)$$

$$\text{F1_Score} = \frac{2 * \text{precision} * \text{recall}}{\text{precision} + \text{recall}} \quad (13)$$

where TP means the true positive rate, FP means the false positive rate, and FN means the false negative rate. Precision presents the percentage of TP in the all-positive of segmentation result, recall denotes the percentage of TP in the all-positive of ground truth, and F1_Score denotes the weighted average of precision and recall.

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TABLE I
EXPERIMENTAL RESULTS OF POTSDAM IRRG TO VAIHINGEN IRRG (%). OVERALL REPRESENTS THE MEAN VALUE OF THE SIX CATEGORIES

Potsdam IRRG → Vaihingen IRRG								
method		clutter /background	impervious surfaces	car	tree	low vegetation	building	Overall
Source only	F1_Score	2.45	43.90	33.51	68.39	42.00	66.05	42.72
	IoU	1.24	28.12	20.13	51.96	26.58	49.31	29.56
AdaptSegNet	F1_Score	8.63	69.08	27.04	69.71	44.16	71.96	48.43
	IoU	4.52	52.76	15.63	53.50	28.34	56.2	35.16
ADVENT	F1_Score	9.52	70.06	33.98	69.88	40.41	73.04	49.48
	IoU	5.00	53.92	20.47	53.71	25.31	57.54	35.99
CLAN	F1_Score	13.93	65.31	29.06	70.86	45.13	71.36	49.27
	IoU	7.48	48.49	17.00	54.87	29.14	55.47	35.41
DAST	F1_Score	1.97	76.80	33.57	**72.06**	51.64	80.41	52.74
	IoU	0.99	62.33	20.17	**56.32**	34.80	67.25	40.31
RCA-DD	F1_Score	6.32	73.04	30.25	68.89	43.27	73.73	49.25
	IoU	3.27	57.53	17.82	52.54	27.61	58.40	36.20
Ours	F1_Score	**51.12**	**79.30**	**54.97**	70.35	**55.90**	**88.30**	**66.66**
	IoU	**34.34**	**65.71**	**37.90**	54.26	**38.80**	**79.05**	**51.68**

IoU is an evaluation metric that measures the percent overlap between predicted and ground-truth segmentation masks and is formulated as follows:

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FN} + \text{FP}}. \tag{14}$$

D. Baselines

To examine the performance of the proposed framework, five baseline methods are compared, including the "source only" method and four state-of-the-art UDA methods, namely, AdaptSegNet [25], adversarial entropy network (ADVENT) [26], category-level adversarial network (CLAN) [30], discriminator attention and self-training network (DAST) [28] and region and category adaptive domain discriminator (RCA-DD) [35]. The "source-only" method trains the source domain on the same Deeplabv2 segmentation network as the proposed framework and then predicts the segmentation results on the target domain. AdaptSegNet employs a multilevel adversarial strategy to narrow the domain gap at multiple levels. ADVENT combines entropy and adversarial learning to improve prediction performance. CLAN introduces category-level adversarial learning to facilitate local alignment. DAST uses discriminator attention and self-training strategies to reduce domain shift. RCA-DD employs a region and category adaptive discriminator to align the distributions of categories.

E. Results

Two groups of experiments, namely, small and large domain shifts, are designed to examine the performance of the proposed method. The two groups of experiments mainly differ in the channels of RSIs used. For the group of small domain shift experiments, IRRG images are used in the source domain and target domain to validate the model performance under geographical environment differences. For the group of large domain shift experiments, RGB images in the Potsdam dataset and the IRRG images in the Vaihingen dataset are used, which means that the datasets of the source and target domains are different in geographical environments and channels of RSIs.

In the training, the labeled images in the source domain and unlabeled images in the target domain are involved. For the testing, the labeled images in the target domain are used for validating the performance of the method. In each group of experiments, the two datasets are exchanged as the source domain and the target domain for two experiments.

1) Small Domain Shift From Potsdam IRRG to Vaihingen IRRG: In this experiment, the Potsdam IRRG dataset and the Vaihingen dataset are used as the source domain and the target domain, respectively. The experimental results are reported in Table I. The proposed method achieves the best overall performance with 66.66% on mean F1_Score and 51.68% on mean IoU (mIoU). Specifically, the mIoU and the mean F1_Score of the proposed method are superior to those of the "source only" method by 22.12% and 23.94%, respectively. Compared with the five UDA-based baselines, the proposed method achieves the best accuracy in five categories of "clutter," "building," "impervious surfaces," "car," and "low vegetation." Furthermore, the accuracy of the "clutter" category with large intraclass variation is greatly improved compared with the other methods. The results suggest that the proposed method combining DSM and RSI is indeed beneficial in breaking the limitations of unimodal UDA methods and improving the accuracy of pixel classification.

To understand the results further, several examples are illustrated in Fig. 6. In Fig. 6(d), the classification results of the "source only" method are highly noisy. In the case of unaligned data distribution, "building," "impervious surfaces," "car," and "clutter" are entangled with one another due to the inconsistency of building styles and the large intraclass variation of clutter. Compared with the "source only" method, the UDA-based baselines, especially DAST and RCA-DD, improve the contextual relationships and reduce the entanglement of different categories. However, their boundary regions are still prone to noise and misclassification. The segmentation results of the proposed method offer less noise and clearer boundaries. With the support of DSM and RSI, the misclassification of "building" and "impervious surfaces" types is reduced evidently. Taking advantage of the auxiliary

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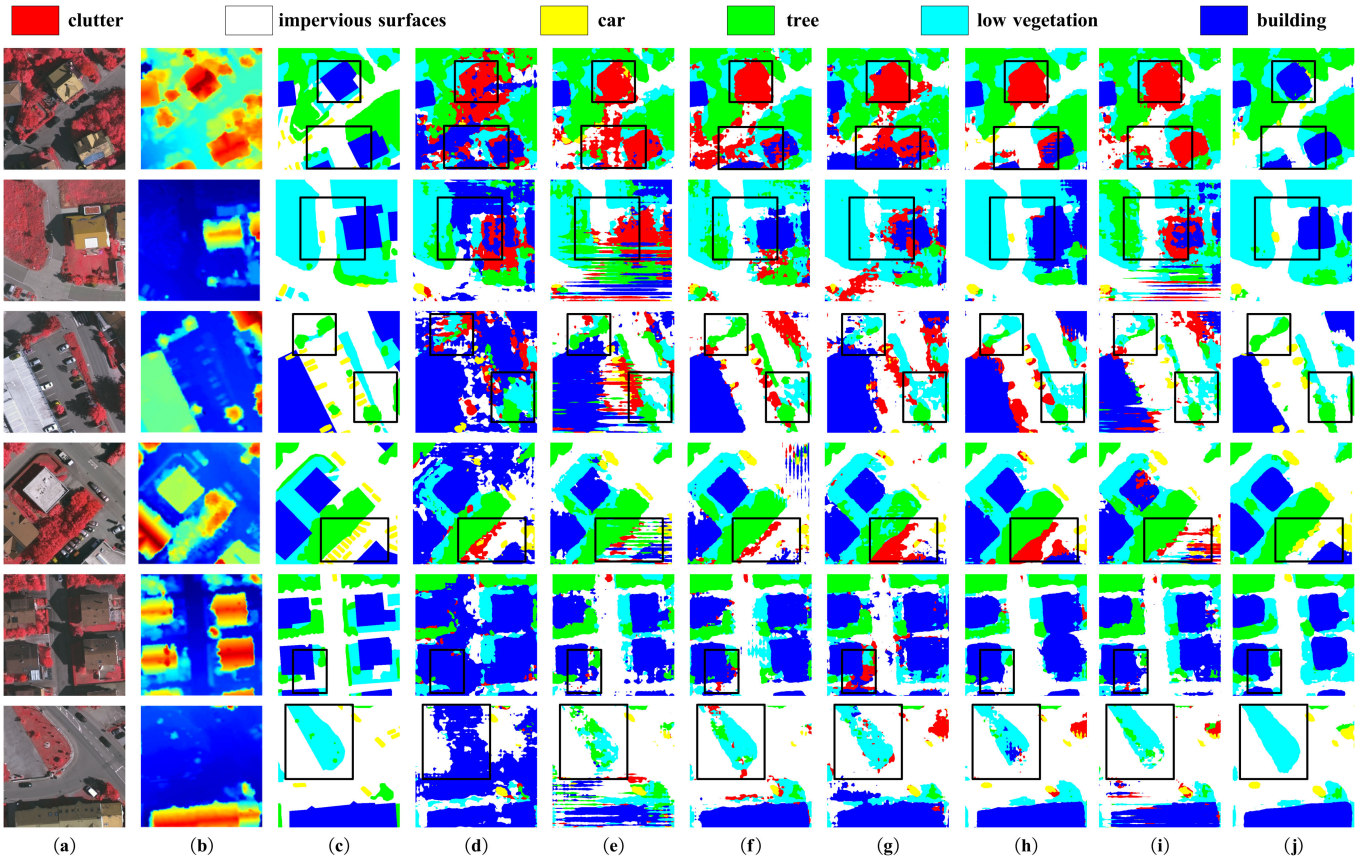


Fig. 6. Segmentation visualization results of different methods in the Potsdam IRRG to Vaihingen IRRG experiment. (a) Image. (b) DSM. (c) GT. (d) Source only. (e) ADVENT. (f) AdapSegNet. (g) CLAN. (h) DAST. (i) RCA-DD. (j) Ours.

discrimination provided by the height information, the proposed method substantially reduces the noise in the shadow regions, as shown in the second and fourth rows. Moreover, misclassification on the boundaries is greatly reduced, as shown in the first and sixth rows. Furthermore, our method more correctly identifies "tree" and "low vegetation," as shown in the third row. These results indicate that the proposed method has superior performance compared with the unimodal UDA methods. To sum up, the method enhances the recognition of certain objects, improves the accuracy of boundary classification, and reduces the interference caused by shadow areas.

2) Small Domain Shift From Vaihingen IRRG to Potsdam IRRG: In this experiment, all methods are trained with the Vaihingen IRRG dataset as the source domain and the Potsdam IRRG dataset as the target domain. The evaluation results of all methods are shown in Table II. Our values on mIoU and mean F1_Score are higher than those of the "source only" method by 20.55% and 21.58%, respectively. Compared with the five UDA-based baselines, the proposed method still achieves the best accuracy. In terms of mIoU, the proposed method exceeds 13.91%, 10.27%, 9.51%, 5.59%, and 5.30% compared with AdaptSegNet, ADVENT, CLAN, DAST, and RCA-DD. The experimental results suggest that the proposed method effectively improves the accuracy of global prediction and increases the confidence of the prediction results.

The prediction results are visualized in Fig. 7. The segmentation result of the "source only" method in Fig. 7(d) is presented with massive noise and relatively poor prediction of small objects. For small objects, such as "car," the proposed method is more robust in predicting position and size and effectively reduces confusion between adjacent objects. In addition, the third to fourth rows show the five UDA-based baselines make serious misclassification and classify shadow areas as "building," and bear dense noise in their boundary regions. Compared with the baselines, our method greatly reduces the misclassification of "building" and "impervious surfaces" by introducing DSM and is rarely affected by shadow as shown in the first, second, and fifth rows. In the fourth and six rows, the segmentation completeness of "tree" and "low vegetation" in our method is greater than those of the baseline methods.

3) Large Domain Shift From Potsdam RGB to Vaihingen IRRG: The experiment of Potsdam RGB to Vaihingen IRRG is conducted to evaluate the performance of the proposed method when the geographic environment and image channels differ. The experimental results are shown in Table III. Compared with the "source only" method, the mean F1_score and mIoU of ours increase by 25.05% and 21.35%, respectively. The proposed method achieves the best accuracy in four of the six categories: "clutter," "impervious surfaces," "car," and "building." The results show that even though the gap of domain adaptation has increased from a small domain to a

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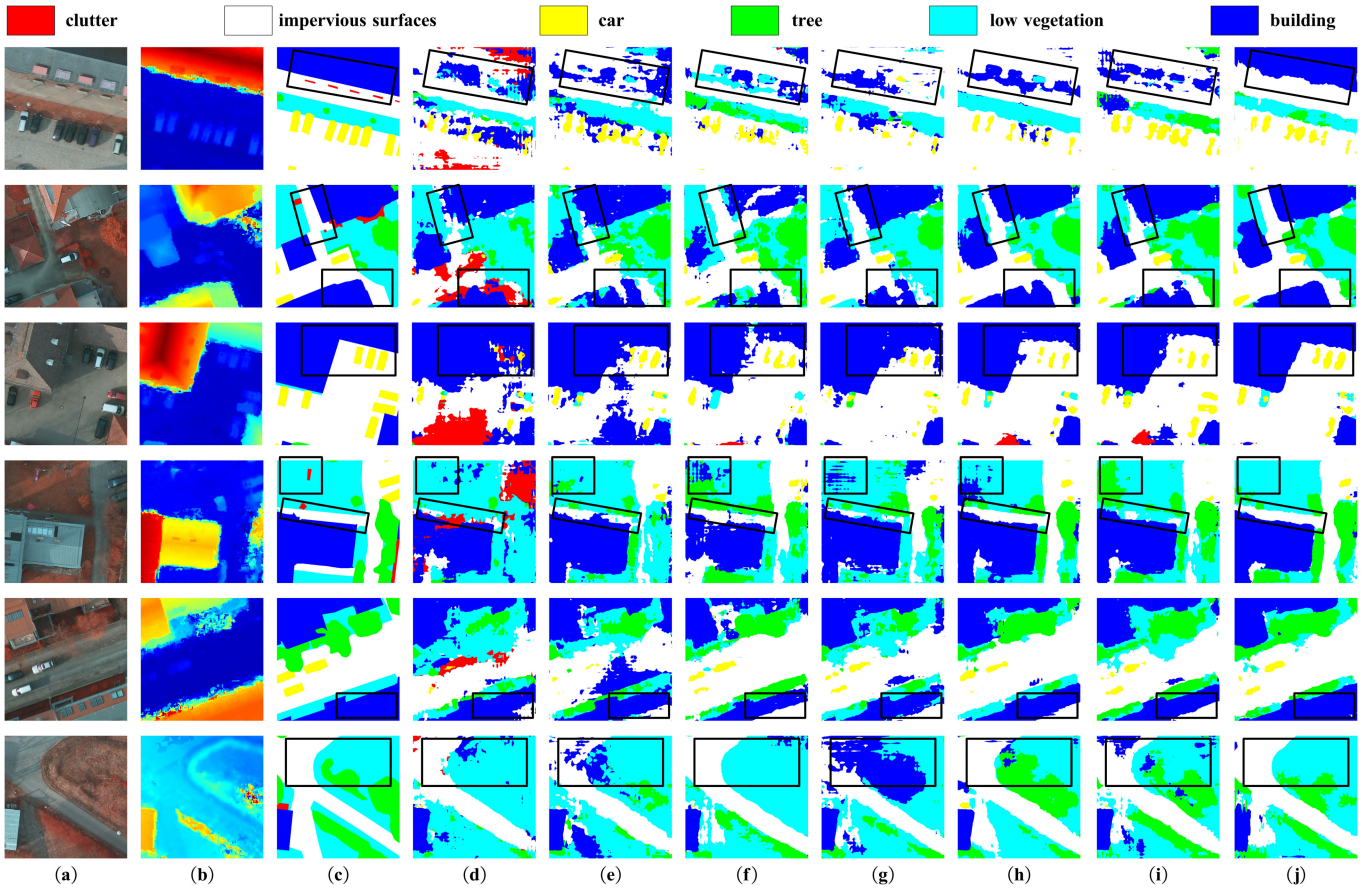


Fig. 7. Segmentation visualization results of different methods in the Vaihingen IRRG to Potsdam IRRG experiment. (a) Image. (b) DSM. (c) GT. (d) Source only. (e) ADVENT. (f) AdapSegNet. (g) CLAN. (h) DAST. (i) RCA-DD. (j) Ours.

TABLE II
EXPERIMENTAL RESULTS OF VAIHINGEN IRRG TO POTSDAM IRRG (%)

Vaihingen IRRG → Potsdam IRRG								
method		clutter /background	impervious surfaces	car	tree	low vegetation	building	Overall
Source only	F1_Score	12.03	52.54	36.33	16.80	59.04	63.93	40.11
	IoU	6.40	35.63	22.20	9.17	41.89	46.98	27.04
AdaptSegNet	F1_Score	22.01	65.97	50.63	35.43	56.73	62.51	48.88
	IoU	12.37	49.22	33.90	21.53	39.60	45.46	33.68
ADVENT	F1_Score	21.0	65.22	56.85	42.45	60.68	69.18	52.66
	IoU	12.10	48.40	39.72	27.26	43.56	52.88	37.32
CLAN	F1_Score	23.34	67.51	56.29	40.64	63.38	69.52	53.45
	IoU	13.21	50.95	39.17	25.50	46.39	53.27	38.08
DAST	F1_Score	23.11	72.93	58.75	51.10	64.61	73.33	57.30
	IoU	13.07	57.40	41.59	34.32	47.73	57.90	42.00
RCA-DD	F1_Score	22.60	71.90	62.50	51.23	60.44	76.27	57.49
	IoU	12.74	56.13	45.46	34.43	43.31	61.65	42.29
Ours	F1_Score	**23.88**	**78.60**	**64.00**	**51.25**	**65.22**	**87.22**	**61.69**
	IoU	**13.56**	**64.75**	**47.06**	**34.45**	**48.39**	**77.33**	**47.59**

large domain, the addition of DSM is still effective. The value of F1_Score and IoU of "tree" and "low vegetation" types are lower than the UDA-based baselines because the RSI feature is sensitive to increasing domain gaps, which indirectly affects the judgment of DSM feature on "tree" and "low vegetation." In addition, the height of the two categories of ground objects does not substantially differ in some areas, resulting in some misclassification between these objects.

The visualization results of the experiment are listed in Fig. 8. In Fig. 8(d), considerably large areas of misclassification are observed with the "source only" method, that is, many pixels are classified as clutter, resulting in producing unreliable

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TABLE III
EXPERIMENTAL RESULTS OF POTSDAM RGB TO VAIHINGEN IRRG (%)

Potsdam RGB → Vaihingen IRRG								
method		clutter /background	impervious surfaces	car	tree	low vegetation	building	Overall
Source only	F1_Score	1.99	51.76	24.71	58.24	14.90	65.16	36.13
	IoU	1.00	34.92	14.1	41.08	8.05	48.32	24.58
AdaptSegNet	F1_Score	2.17	70.27	34.18	**72.23**	34.71	68.82	47.06
	IoU	1.10	54.16	20.61	**56.53**	21.00	52.46	34.31
ADVENT	F1_Score	4.41	68.55	29.48	70.00	33.95	64.44	45.14
	IoU	2.26	52.14	17.29	53.84	20.44	47.53	32.25
CLAN	F1_Score	5.39	63.58	24.93	69.12	31.25	63.87	43.02
	IoU	2.77	46.61	14.24	52.81	18.52	46.92	30.31
DAST	F1_Score	3.10	71.39	34.11	71.77	**38.90**	79.20	49.75
	IoU	1.58	55.52	20.57	55.97	**24.15**	65.56	37.22
RCA-DD	F1_Score	0.69	67.78	29.57	69.72	30.10	66.78	44.10
	IoU	0.34	51.27	17.35	53.52	17.71	50.13	31.72
Ours	F1_Score	**48.41**	**75.58**	**55.29**	68.26	37.10	**82.43**	**61.18**
	IoU	**31.94**	**60.75**	**38.21**	51.81	22.78	**70.10**	**45.93**

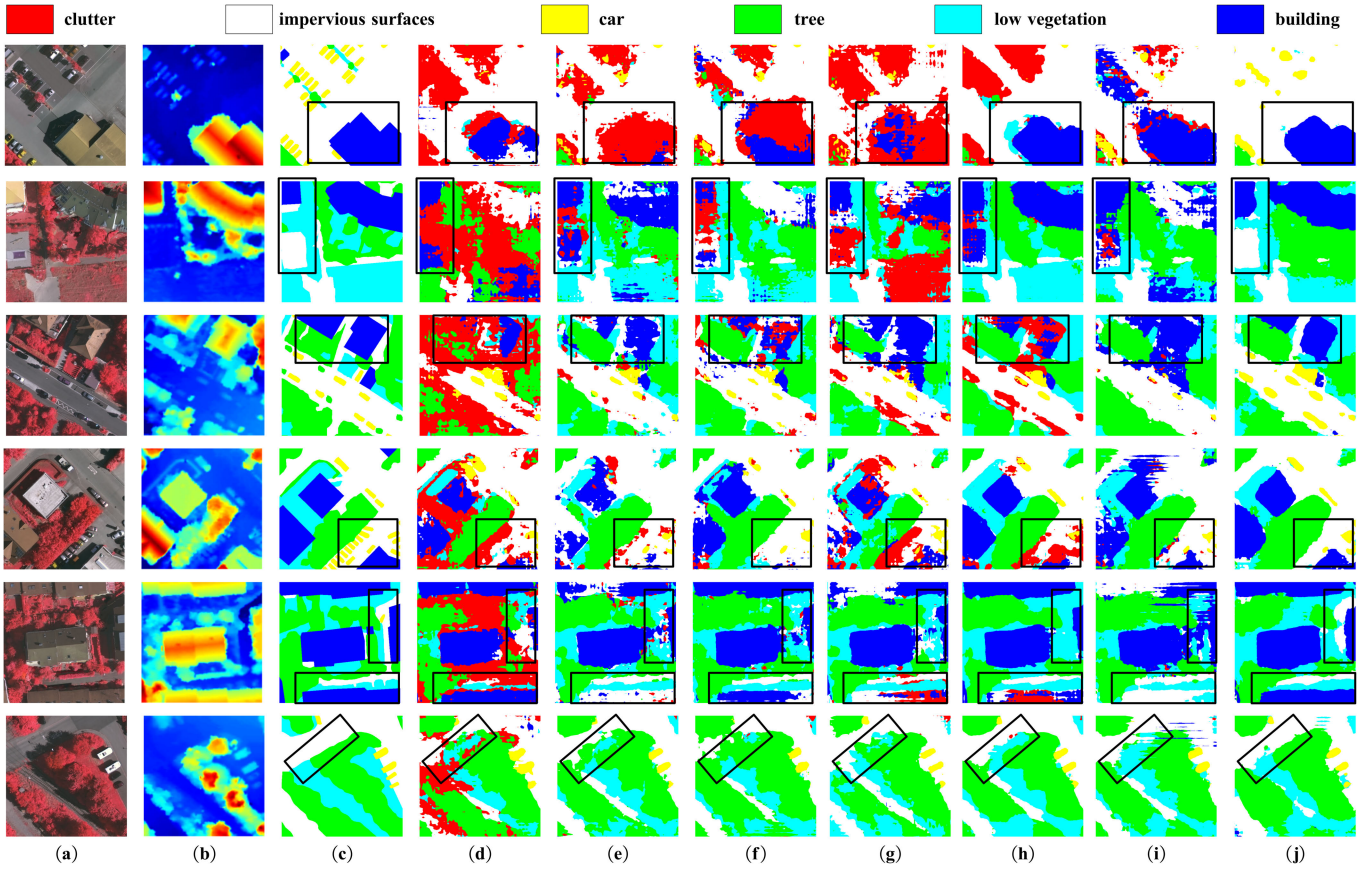


Fig. 8. Segmentation visualization results of different methods in the Potsdam RGB to Vaihingen IRRG experiment. (a) Image. (b) DSM. (c) GT. (d) Source only. (e) ADVENT. (f) AdapSegNet. (g) CLAN. (h) DAST. (i) RCA-DD. (j) Ours.

results. In the first to fifth rows of Fig. 8, "clutter" in Fig. 8(e)–8(i) is confused with the other categories in various degrees, resulting in a sharp decrease in the accuracy of the "clutter" categories. In addition, as shown in the second row, similar spectral regions are easily confused and other methods tend to predict "impervious surfaces" as "building" or "clutter." The proposed method effectively reduces misclassification on confusing areas with the combination of data from two different perspectives. Specifically, this method directly and remarkably improves the segmentation accuracy of "clutter," "building," and "impervious surfaces" categories. In addition, MDDANet improves area boundaries, especially in the shadow regions, as shown in the fourth to sixth rows. Overall, even though the difficulty of domain adaptation has increased, our method still outperforms other unimodal methods for identifying ground objects.

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TABLE IV
EXPERIMENTAL RESULTS OF VAIHINGEN IRRG TO POTSDAM RGB (%)

Vaihingen IRRG → Potsdam RGB								
method		clutter /background	impervious surfaces	car	tree	low vegetation	building	Overall
Source only	F1_Score	19.94	59.72	49.95	20.94	37.44	58.08	41.01
	IoU	11.07	42.57	33.28	11.69	23.04	40.92	27.10
AdaptSegNet	F1_Score	24.23	60.86	53.87	40.95	37.08	64.57	46.93
	IoU	13.79	43.74	36.87	25.75	22.76	47.68	31.76
ADVENT	F1_Score	18.58	59.27	54.27	45.98	**43.06**	66.52	47.95
	IoU	10.24	42.12	37.24	29.85	**27.44**	49.84	32.79
CLAN	F1_Score	19.78	62.67	53.98	35.99	42.51	56.54	45.25
	IoU	10.98	45.64	36.97	21.94	26.99	39.42	30.32
DAST	F1_Score	20.51	66.22	56.34	**58.50**	12.66	61.84	46.01
	IoU	11.42	49.51	39.23	**41.34**	6.75	44.76	32.17
RCA-DD	F1_Score	22.82	66.36	56.82	48.63	22.63	73.73	48.50
	IoU	12.88	49.65	39.68	32.13	12.76	58.39	34.25
Ours	F1_Score	**25.15**	**74.64**	**66.13**	51.56	34.65	**81.07**	**55.53**
	IoU	**14.39**	**59.54**	**49.40**	34.73	20.96	**68.16**	**41.20**

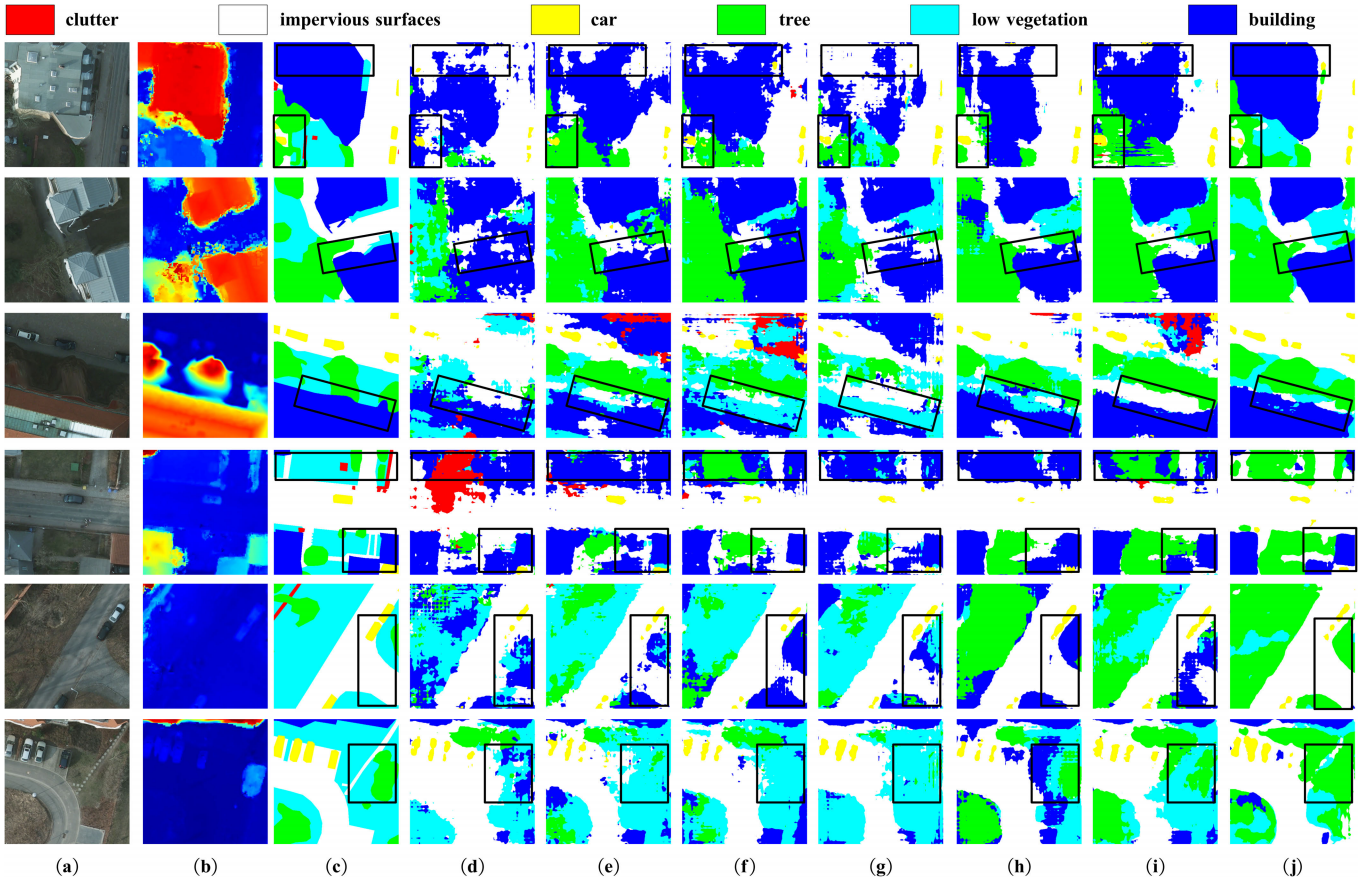


Fig. 9. Segmentation visualization results of different methods in the Vaihingen IRRG to Potsdam RGB experiment. (a) Image. (b) DSM. (c) GT. (d) Source only. (e) ADVENT. (f) AdapSegNet. (g) CLAN. (h) DAST. (i) RCA-DD. (j) Ours.

4) Large Domain Shift From Vaihingen IRRG to Potsdam RGB: The experimental results with Vaihingen IRRG as the source domain and Potsdam RGB as the target domain are shown in Table IV. Compared with the "source only" method, the proposed method improves the overall mIoU and F1_Score by 14.1% and 14.52%, respectively. A large improvement is achieved in four categories. Especially, "building" and "impervious surfaces" are far superior to baseline methods. The downside is that our method performs poorly in discriminating "tree" and "low vegetation," which may be because the DSM of the two categories is very similar in some regions.

The detailed segmentation results are illustrated in Fig. 9. The "source only" method, shown in Fig. 9(d), is not very reliable for the prediction of "low vegetation" and "tree," and

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TABLE V
ABLATION STUDY FOR POTSDAM IRRG TO VAIHINGEN IRRG TASK

Method	MPE	MTD	mean F1_Score (%)	mIoU (%)
baseline			40.80	31.21
with MPE	✓		54.16	40.70
with MTD		✓	57.77	44.73
with MPE, MTD	✓	✓	59.93	45.99

TABLE VI
ABLATION RESULTS FOR UDA MODULE IN POTSDAM IRRG TO VAIHINGEN IRRG TASK

Method	UDA	mean F1_Score (%)	mIoU (%)
MDDANet (w/o UDA, RPF)		45.79	32.96
MDDANet (w/o RPF)	✓	59.93	45.99

many cavernous areas appear inside the buildings. In the first, third, and sixth rows, the UDA-based baselines improve the prediction of "low vegetation" and "tree," but their results also suffer from some misclassifications. Instead, our method accurately predicts the adjacent object boundaries between these two and other categories. Moreover, our method separates "building" from "impervious surfaces" easily. In particular, our method reduces the substantial misclassification and omission of buildings because the height of the buildings differs from that of other objects. Thus, the introduction of DSM enhances the ability to identify building areas. In general, the proposed method can improve the segmentation integrity of ground objects in various situations, which shows that our method is robust. These results further prove that DSM can be generally used to improve the segmentation accuracy of RSI.

V. DISCUSSION

A. Ablation Study

1) Effect of Multimodal Fusion Module: The effectiveness of two key modules in the proposed method, namely, MPE and MTD, is examined in this section. The network with the encoder of DSM and ASPP added based on the "source only" method is adopted as a baseline and then each of the components, MPE and MTD, is added progressively. Considering more application scenarios with small domain migration, small domain shift experiments from Potsdam IRRG to Vaihingen IRRG images are conducted in this section.

Table V shows the results of different ablation experiments. The "with MPE" improves the baseline from 40.80% to 54.16% and 31.21% to 40.70% in terms of mean F1_Score and mIoU, respectively. The improvement demonstrates that the introduction of MPE is effective for enhancing the segmentation performance of the model. It is probably due to the well-learning DSM further bringing more support information for the prediction ground objects. The "with MTD" brings mean F1_Score and mIoU further improvements of 16.97% and 13.52%, respectively. The improvement indicates that the filtering block and fusion block in MTD can enhance the recognition ability of ground objects and increase the accuracy rate. The "with MPE, MTD" integrating two modules improves the segmentation accuracy and achieves the best mIoU of 45.99% and the best mean F1_Score of 59.95%.

To illustrate the contribution of different modules more intuitively, the prediction results and the feature distribution of the three images using t-SNE are visualized, as shown in Fig. 10. Fig. 10(c1)–10(f1) are the prediction results, and Fig. 10(c2)–10(f2) are the t-SNE visualization results. The "w

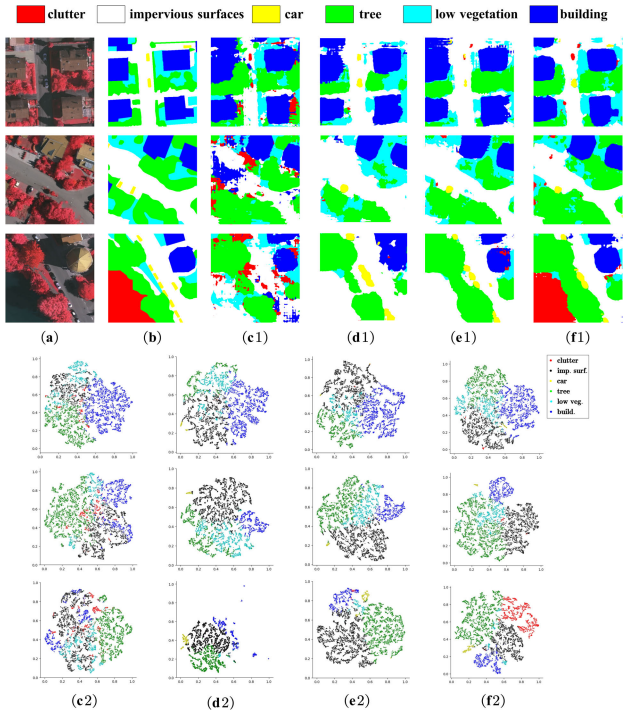


Fig. 10. Some examples of prediction results and t-SNE visualization. (a) RSI. (b) GT. (c1) and (c2) Baseline. (d1) and (d2) With MPE. (e1) and (e2) With MTD. (f1) and (f2) With MPE, MTD.

MPE" reduces the appearance of noise and provides better prediction of small objects, and "w MTD" further regularizes the shape of the ground objects and increases the separability of classes. The model with "MPE" and "MTD" achieves the best prediction results.

2) Effect of UDA: To verify the effectiveness of the UDA module, the Potsdam IRRG dataset is trained after removing the UDA and RPF module from MDDANet; that is, MDDANet (w/o UDA, RPF) and then the results are tested directly on the target domain. The experimental results and visualized confusion matrix are shown in Table VI and Fig. 11, respectively. The mean F1_Score and mIoU of MDDANet (w/o RPF) are 14.14% and 13.03% higher than those of MDDANet (w/o UDA, RPF), respectively. After adding UDA to the multimodal model, the source and target domains are globally aligned. In Fig. 11, the accuracies are dramatically improved for most categories except the type of "tree." While the TP of "tree" decreases, the cases of other categories being misclassified as "tree" are greatly reduced, thereby adding UDA favors the prediction of "tree." This finding suggests that global alignment strategies are necessary when transferring the model from the source domain to the target domain.

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TABLE VII
ABLATION RESULTS FOR RPF MODULE IN POTSDAM IRRG TO VAIHINGEN IRRG TASK

Method		clutter /background	impervious surfaces	car	tree	low vegetation	building	Overall
MDDANet (w/o RPF)	F1_Score	19.31	75.84	54.97	69.36	53.07	87.04	59.93
	IoU	10.69	61.08	37.90	53.09	36.12	77.06	45.99
MDDANet	F1_Score	51.12	79.30	54.97	70.35	55.90	88.30	66.66
	IoU	34.34	65.71	37.90	54.26	38.80	79.05	51.68

TABLE VIII
COMPARISON OF INFERENCE TIME OF DIFFERENT UDA SEMANTIC SEGMENTATION METHODS

Method	AdaptSegNet	CLAN	ADVENT	DAST	RCA-DD	MDDANet
Inference time(s)	0.169	0.115	0.154	0.101	0.201	0.183

TABLE IX
QUANTITATIVE RESULTS OF USING ASPP OR AAFI AS DECODER OF DSM

Method		clutter /background	impervious surfaces	car	tree	low vegetation	building	Overall
ASPP	F1_Score	17.29	76.72	52.10	66.46	51.88	81.00	57.58
	IoU	9.47	62.24	35.22	49.76	35.03	68.07	43.30
AAFI	F1_Score	19.31	75.84	54.97	69.36	53.07	87.04	59.93
	IoU	10.69	61.08	37.90	53.09	36.12	77.06	45.99

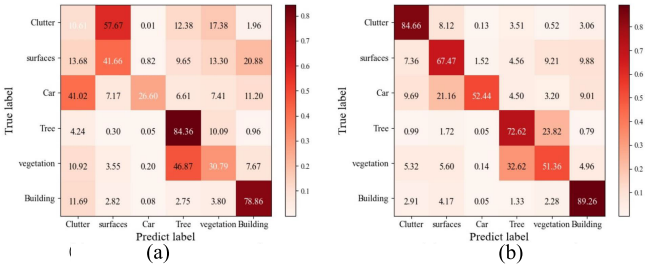


Fig. 11. Results of confusion matrix visualization with and without domain adaptation. (a) MDDANet (w/o UDA, RPF). (b) MDDANet (w/o RPF).

3) Effect of RPF: To verify the validity of the RPF module in the inference phase, the quantitative results are presented in Table VII. The threshold μ is set to different values in RPF to achieve the best results. With RPF, the mean F1_Score and mIoU improve by 6.72% and 6.69%, respectively. Especially, the greatest improvement is obtained on "clutter" and "impervious surfaces" because RPF further rectifies the entanglement of "clutter" and "impervious surfaces." The IoU of the four other categories also generally increases by about 2%. These facts adequately demonstrate that RPF contributes to capturing fine details, thus facilitating the improvement in the performance of pixel-level classification.

B. Time Cost Analysis

To examine the time cost of the proposed method in the inference phase further, Table VIII shows the average inference time for one image with the UDA-based baselines and our method. Among the five methods, the inference speed of DAST is the fastest. The speed of our method is about 0.082 s slower than that of DAST, but the accuracy of our method is much higher than these unimodal methods. For example, in the Potsdam IRRG to Vaihingen IRRG experiment, MDDANet and DAST obtain 51.68% and 40.31% mIoU, respectively. Paying a certain amount of extra time to gain an improvement in accuracy is acceptable.

C. Effect of AAFI

In the MTD module, the proposed model employs the AAFI block to focus on the local area and filter the noisy information. To evaluate the effect of AAFI, the AAFI block is replaced with a classical block, ASPP, which is the same as the decoder of RSI. The experimental results and confusion matrices are reported in Table IX and Fig. 12. Table IX shows the overall accuracy is improved by the introduction of AAFI because the AAFI provides more refined classification information, that is, AAFI filters out some incorrect prediction results. Fig. 12 shows that AAFI strongly reduces confusing classifications of "clutter" and "building," and greatly increases the TP of "building," making the IoU of "building" 8.99% higher than ASPP. Although the increase in the FN of impervious surfaces leads to a slightly lower accuracy obtained by AAFI than ASPP, its FP is reduced. This finding means that our method reduces the cases where other classes are misclassified as impervious surfaces. Comparing the prediction results for all categories, AAFI is far superior to ASPP. This result shows that AAFI is more applicable to our framework.

D. Effect of Region Selection in RPF

RPF enhances model prediction by fusing two probability maps, where Wb is used for the selection of the fusion

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TABLE X
QUANTITATIVE RESULTS OF FOUR GROUPS OF *Wb* IN RPF

Discarded categories	car	clutter	mIoU(%)	mean F1_Score (%)
Our (car)	✓		51.68	66.66
car and clutter		✓	46.71	60.52
clutter	✓	✓	46.83	60.51
none			50.91	65.82

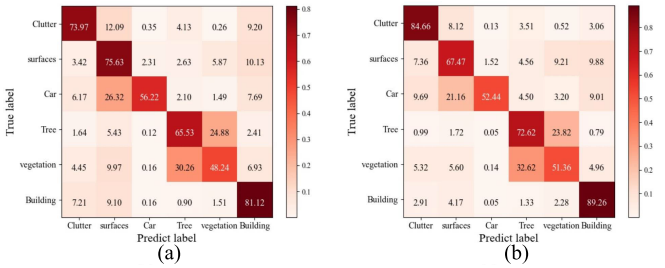


Fig. 12. Confusion matrix visualization using ASPP and AAFI. (a) ASPP. (b) AAFI.

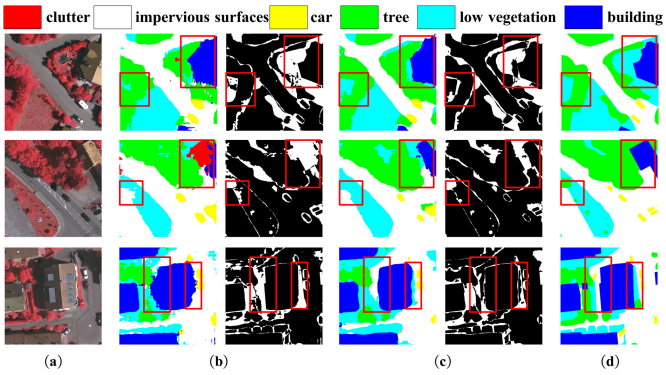


Fig. 13. Visualized prediction results from the RSI and fusion paths. The left parts of (b)–(c) present the prediction results and their right parts present the wrong prediction results (incorrect prediction is in white). (a) Image. (b) RSI. (c) Fusion. (d) GT.

region. The proposed model discards "car" for obtaining *Wb* because of the small intraclass variation and insignificant height values of "car." The effect of the region selection of *Wb* is examined in this section. Given that two categories, "car" and "clutter," are ignored when training the DSM segmentation network, only whether to discard regions predicted as these two categories during fusion are considered. Four groups of experiments are conducted by discarding "car," "clutter," "car" and "clutter," and none.

The mIoU and mean F1_Score of the experimental results are presented in Table X. Our selection on *Wb* achieves the best prediction accuracy, which suggests the fine-filtered DSM probability maps are effective, and discarding areas predicted to be "clutter" result in a degradation of the model performance. Because "clutter" has a large intraclass variability, when discarding "clutter" areas, the function of DSM is weakened, leading to decreased performance.

E. Effect of the Input DSM

To illustrate the effects of the input DSM, we visualized the prediction results of both the RSI path and fusion path for some samples in Fig. 13, and highlighted the improved regions with red rectangular boxes. As shown in Fig. 13, some noises and misclassifications on the boundaries from the RSI path were improved with the introduction of DSM, including the buildings presented blurry boundaries (in the first row), misclassification as clutters (in the second row) and overshadowed by buildings (in the third row). Generally, the boundaries of ground objects with distinct height differences are well predicted in the fusion path with less misclassification. The experimental results indicate that the input of DSM helps to improve the discriminability of ground objects, which benefits from DSM providing a domain-invariant feature that helps discriminate objects with different heights.

VI. CONCLUSION

This article proposes a framework to exploit the complementary information of DSM and RSI fully for improving the UDA-based semantic segmentation of high-resolution RSI and address the limitations of unimodal RSI data in the semantic segmentation task. The experiments on the ISPRS 2D Semantic Labeling Contest dataset show that the proposed method can help discriminate for some ground objects with similar spectral features or in shaded regions, such as buildings and roads, low vegetation, and trees, and refine the region boundaries between these objects, improving considerably the performance of domain adaptive semantic segmentation substantially. This study explores the methodology of introducing DSM to UDA semantic segmentation of RSI, which can serve for various semantic segmentation scenarios where ground objects differ in height, such as building extraction, road detection, and so on. Moreover, the proposed framework is a new attempt to integrate multiple types of data that enriches the approach of fusing multimodal data for addressing UDA tasks.

In addition to DSM, many data sources may contribute to the semantic segmentation of RSI. Fusing more data sources for the semantic segmentation of RSI is worth pursuing. In addition, further consideration of the optimization strategies of UDA to improve model performance is needed.

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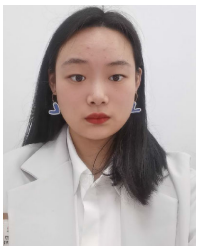
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